

AI-compute supply chain: chart pack

Current state of the analysis. Data: UN Comtrade + TDM monthly panel (60 HS6 codes, 2017-01..2026-04, checked against Atlas) and Atlas annual data. Sections: 1 the chain as flows (2024) -- 2 the chain in stages -- 3 the factor model over time -- 4 concentration & fragmentation (sections 3 and 4 use the AI-compute codes only). Details: docs/data.md, docs/supply-chain-narrative.md, results/.

2026-08-20

1. The chain as flows

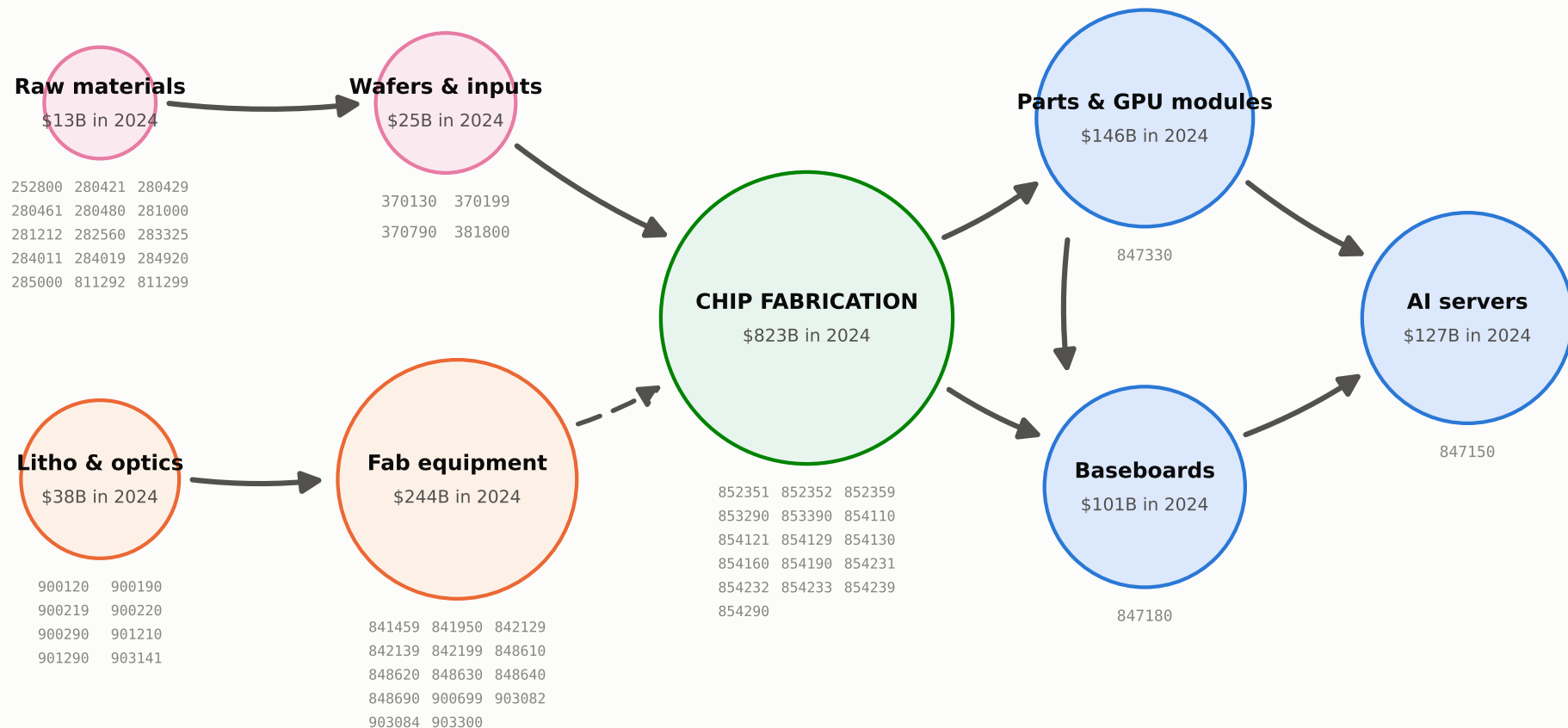
2024 trade flows. China and Hong Kong are counted as one (CHK) throughout. Stages 1-5 use annual Atlas data; stages 6-8 our monthly panel.

The shape of the chain

1. The chain as flows

Two input branches feed chip making; finished hardware flows out to the right. Pink inputs are used up per chip; orange equipment builds capacity. Blue: the three codes of our monthly panel. Circle size grows with the log of 2024 trade.

The AI-compute supply chain — stylized shape, with the HS6 codes in each node



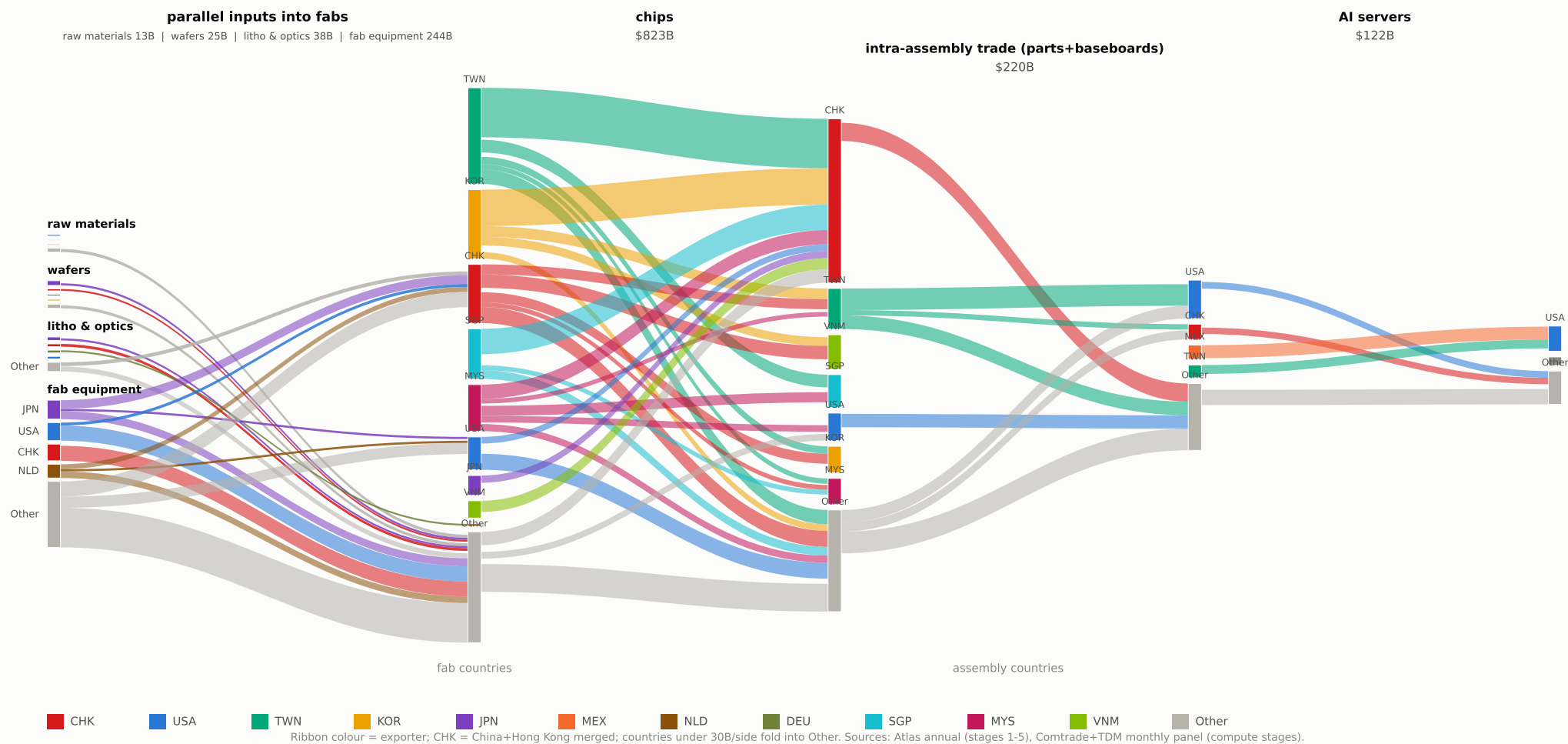
Two input branches, one output line. Materials (pink) -- silicon, chemicals, wafers -- are used up with every chip made (solid arrows). Equipment (orange) -- lithography and other fab tools -- is investment in capacity: in Taiwan and Korea, equipment imports rise six to twelve months before chip exports (dashed arrow). Korean memory and Taiwanese chips are packaged together, usually in Taiwan. Chip design earns as a service and crosses no border. Circle diameters grow with the log of 2024 world trade. Blue marks the three codes of our monthly panel (the Fed AI-compute basket). A few niche inputs (lasers, vacuum pumps and valves, chip substrates) fall outside the 60 codes. Codes: OECD semiconductor mapping + Fed basket (docs/data.md, section 1).

The whole chain on one dollar scale

1. The chain as flows

Band width = 2024 dollars. The chips stage dominates; small input flows are drawn but too thin to see.

The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (one dollar scale — band widths nominally comparable everywhere)



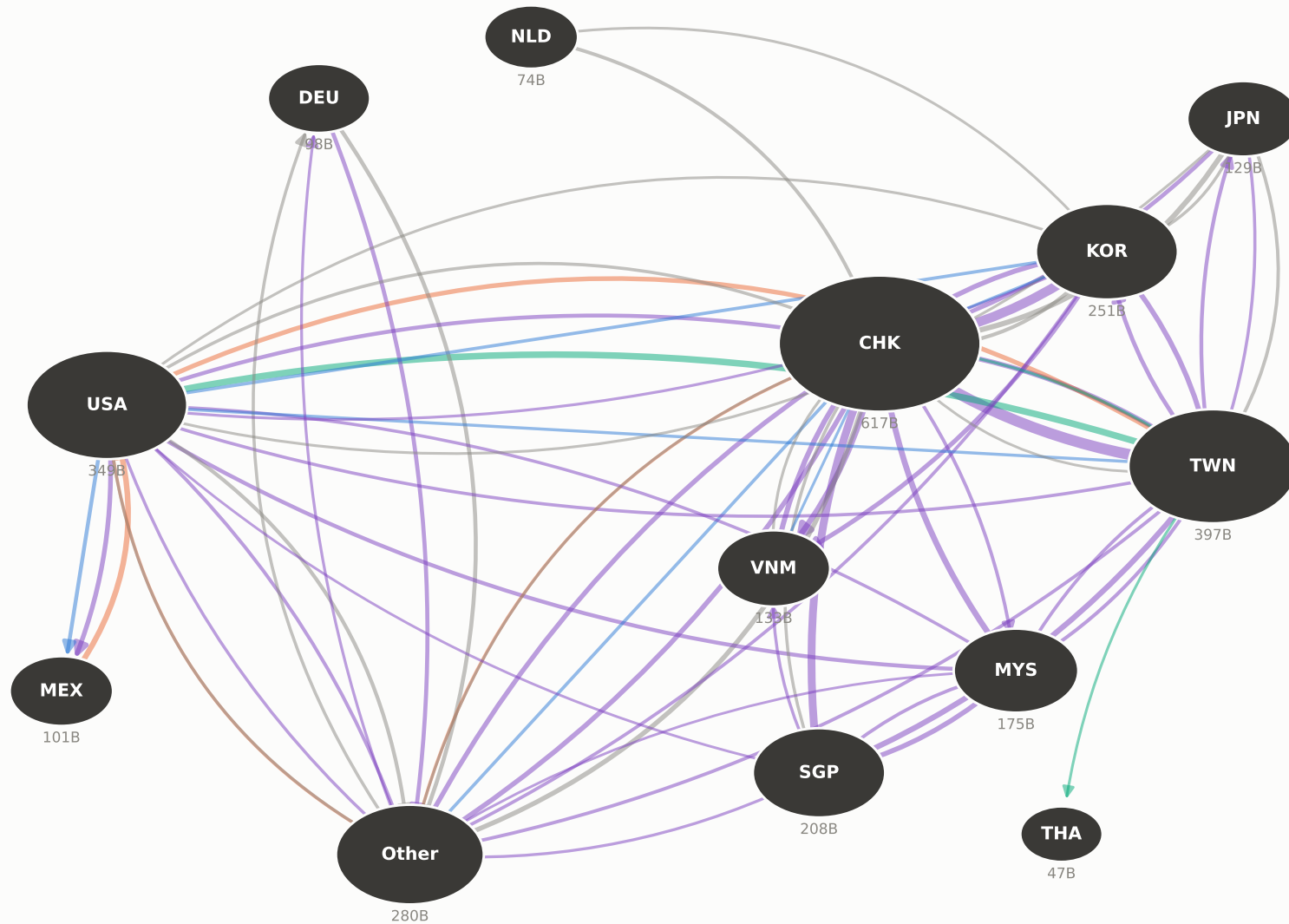
The same flows as a country map

1. The chain as flows

Countries as nodes, flows as edges colored by stage. Shows combined roles: Mexico takes in chips and parts and ships out servers.

The AI-compute supply chain as a country network, 2024 — edges by product stage, width ~ \$

fab inputs chips parts baseboards servers flows >= 5B; node label = total involvement; CHK = China+HK



2. The chain in stages

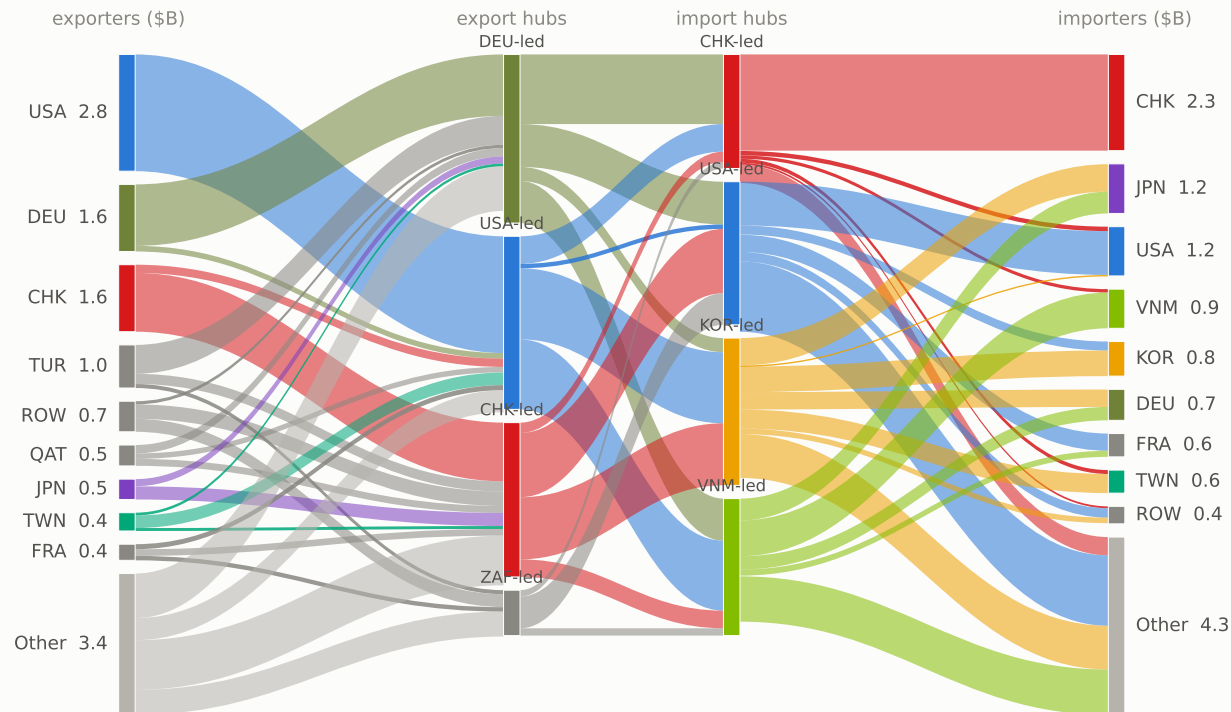
One page per stage: exporters on the left, importers on the right, with the model's groupings in between. Groupings fit the 2024 flows closely (R-squared 0.95-0.99).

Stage 1 -- raw materials

2. The chain in stages

Reading guide for all stage pages: left = exporters, right = importers, middle = the model's export and import groups. Much raw-materials trade is not chip-related (the codes also cover welding gas, abrasives, fertilizer inputs), so flows scatter widely.

Stage 1 — Raw materials (silicon 280461, rare gases, Ga/Ge, chemicals) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$13B



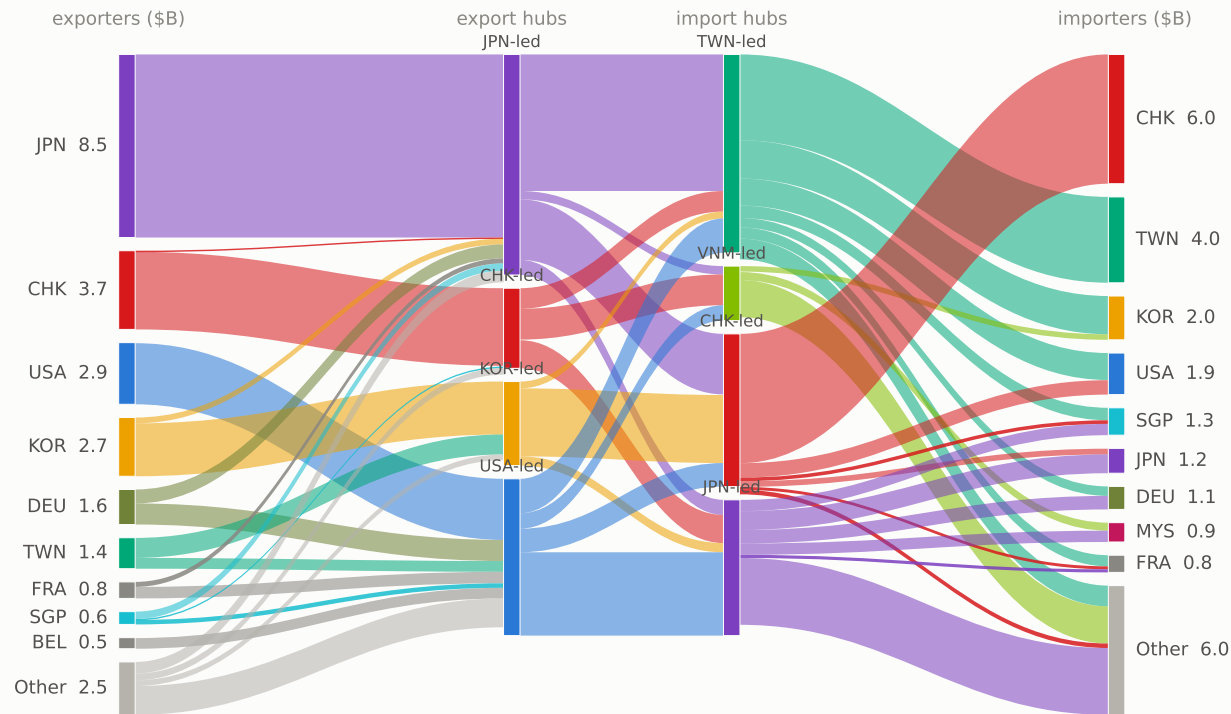
Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

Stage 2 -- wafers and wafer inputs

2. The chain in stages

Japan dominates the export side.

Stage 2 — Wafers & wafer inputs (381800, 3701xx/370790) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$25B

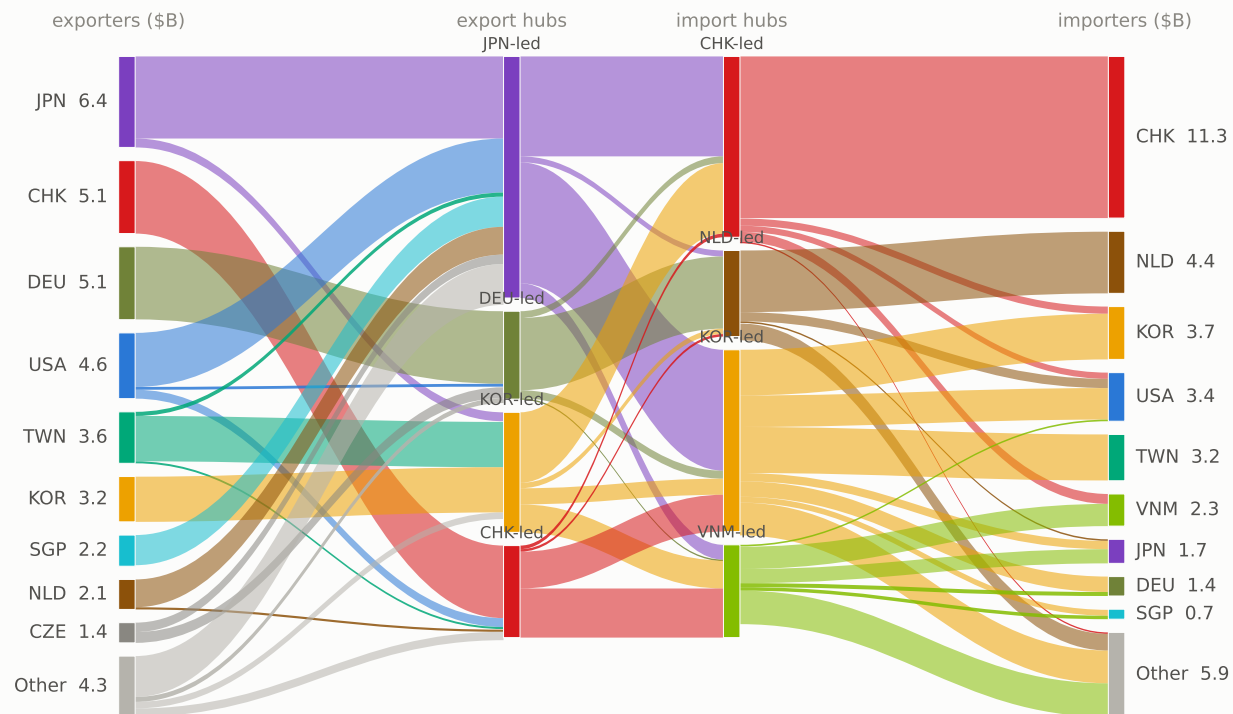


Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

Stage 3 -- lithography and optics inputs

2. The chain in stages

Stage 3 — Lithography & optics inputs (9001xx/9002xx, 901210/90, 903141) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$38B



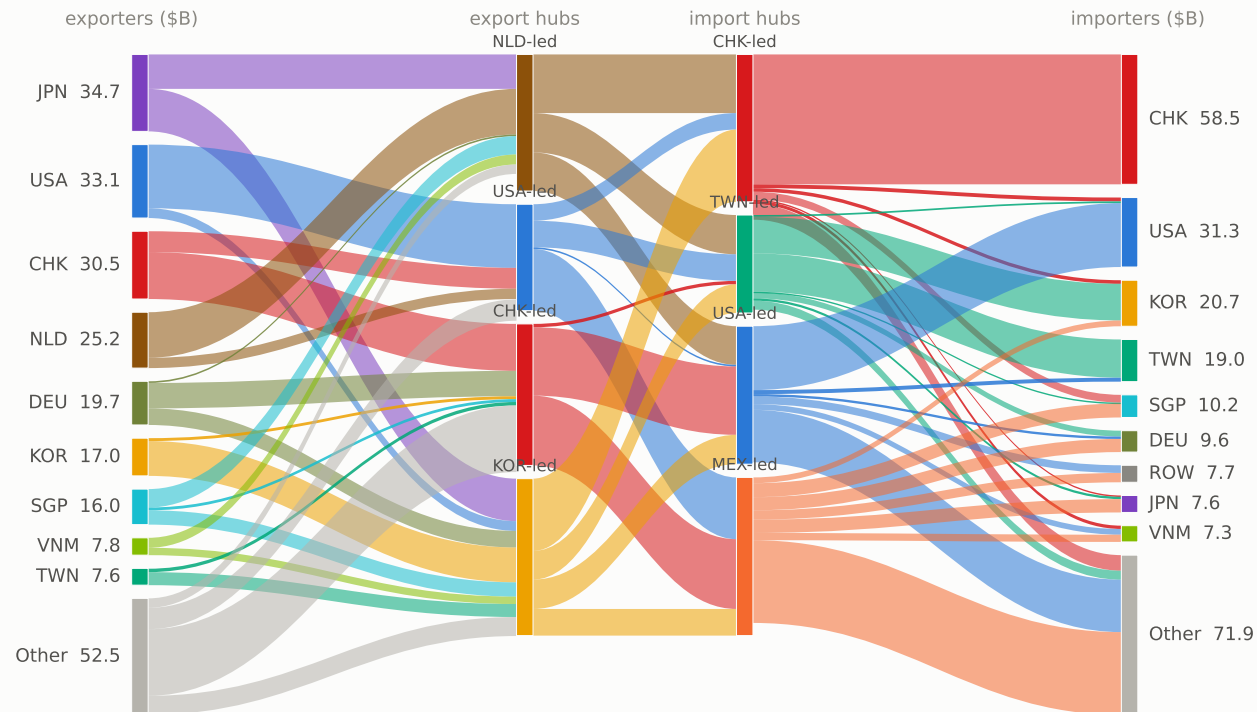
Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

Stage 4 -- fab equipment

2. The chain in stages

Machines, not materials: countries buy these to build capacity, and their imports rise 6-12 months before chip exports.

Stage 4 — Fab equipment (8486xx, metrology, fab plant) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$244B



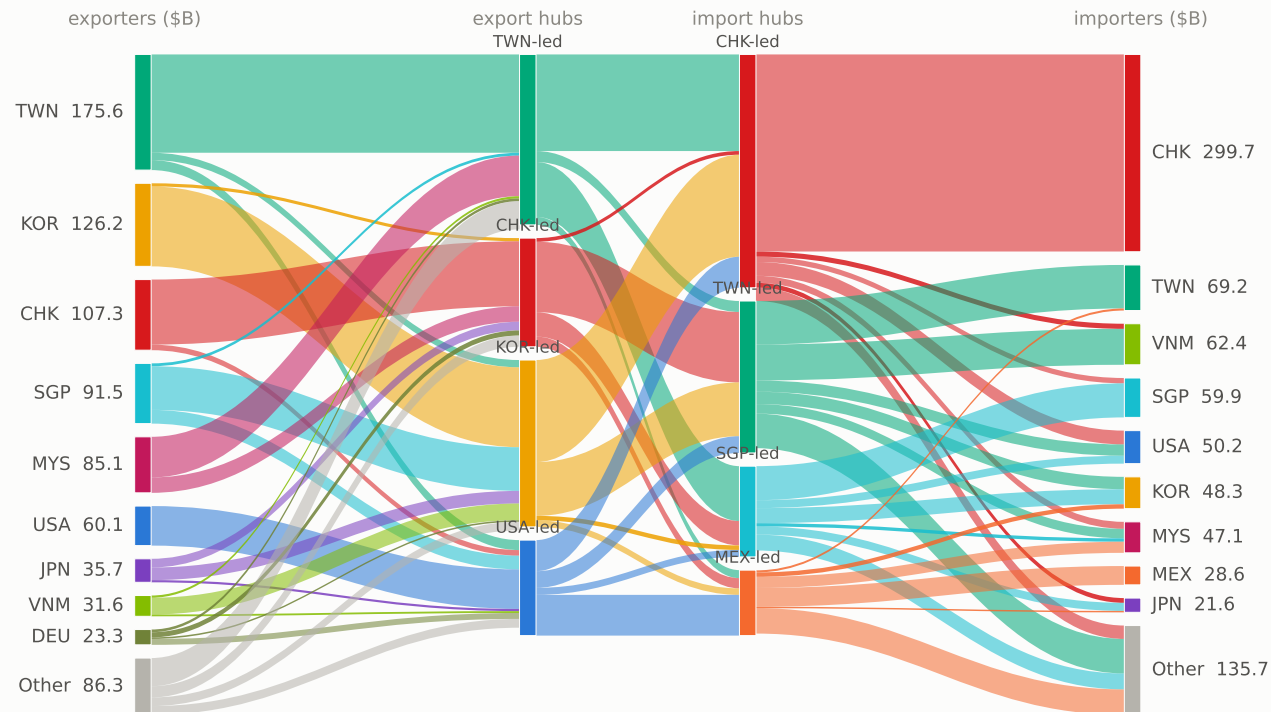
Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

Stage 5 -- chips

2. The chain in stages

The biggest stage, and the whole electronics industry -- phones, cars and PCs, not just AI. Most chips flow to Asian assembly; the China sphere absorbs about 43%.

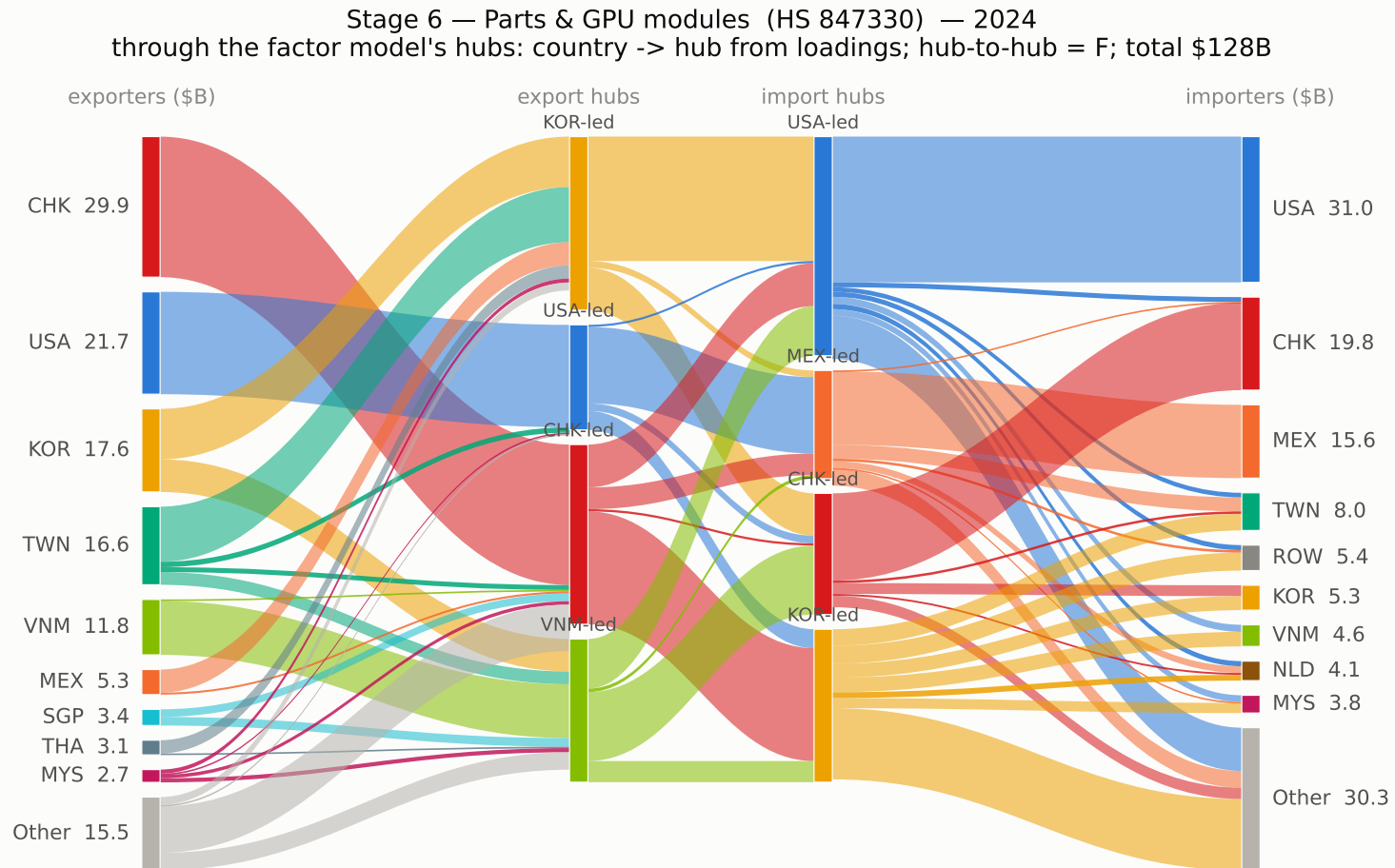
Stage 5 — Chips (logic, memory, discretes) (8542xx, 8541xx, media) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$823B



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 (no monthly panel upstream)); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral.

Stage 6 -- parts and GPU modules (847330)

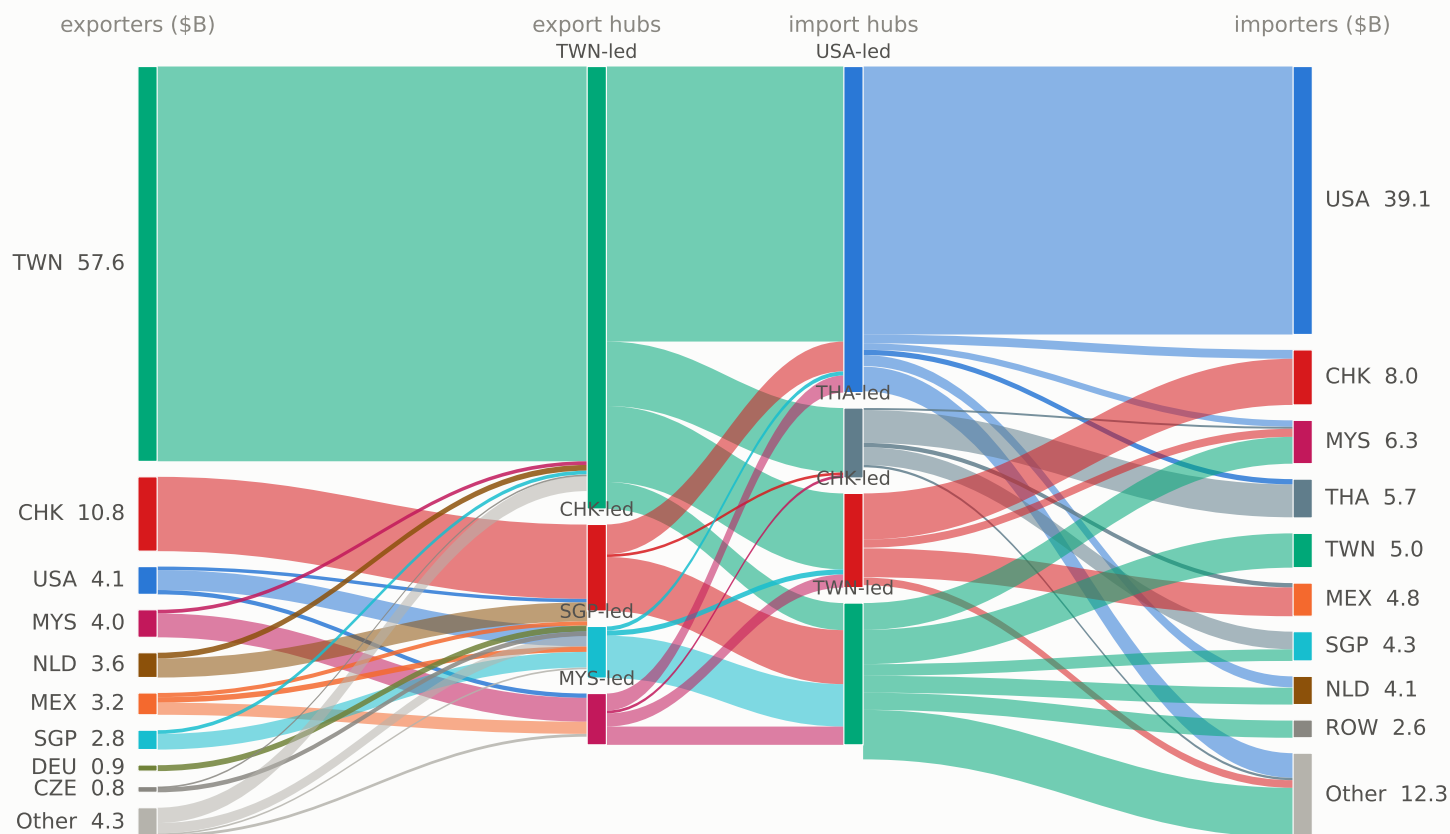
2. The chain in stages



Stage 7 -- baseboards (847180)

2. The chain in stages

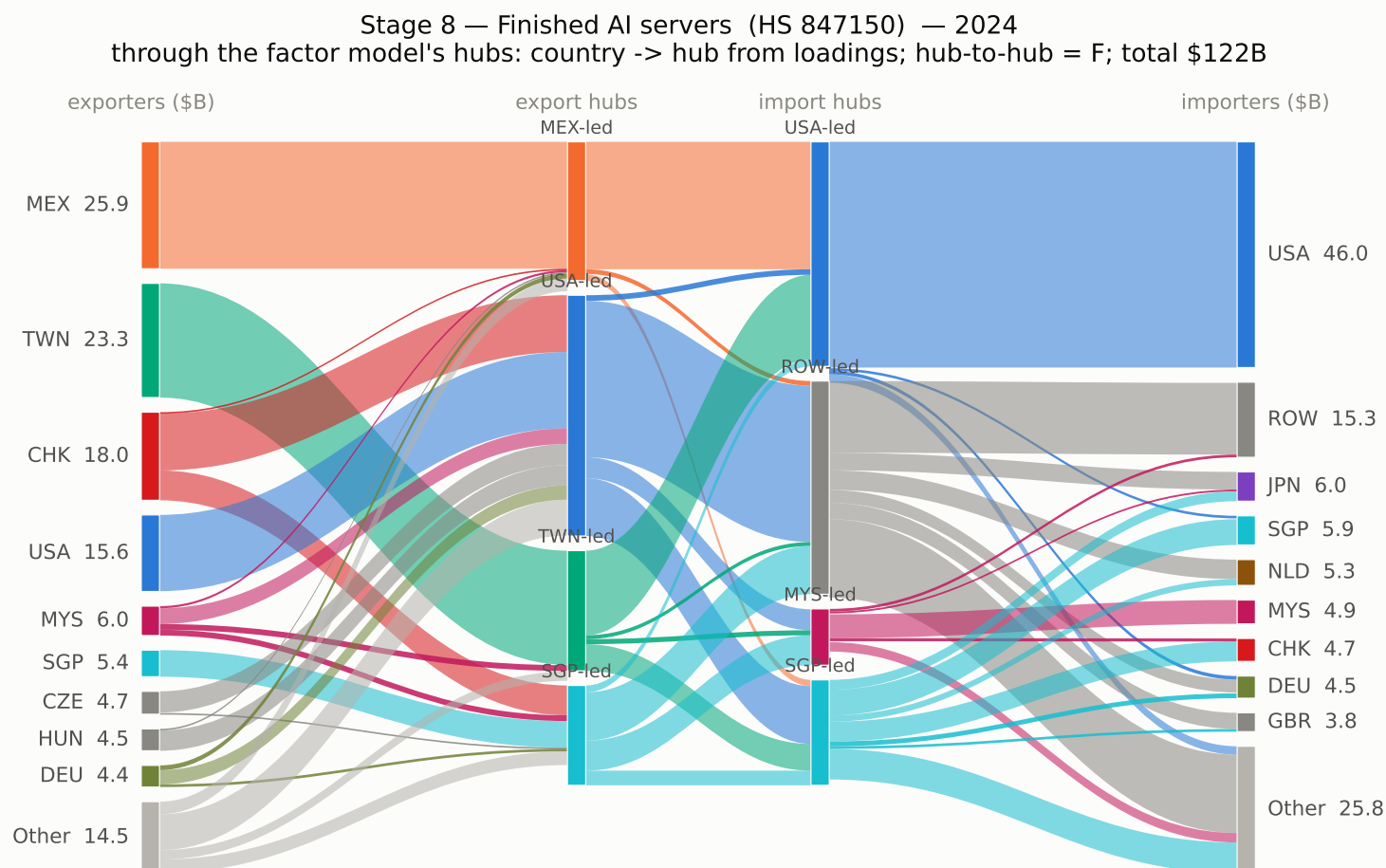
Stage 7 — Baseboards / other units (HS 847180) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$92B



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

Stage 8 -- AI servers (847150)

2. The chain in stages



Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); pooled 2024 monthly MFM); outer columns rescaled to actual totals. CHK = China+HK. Data: Comtrade+TDM monthly panel.

3. The factor model over time -- AI-compute codes only

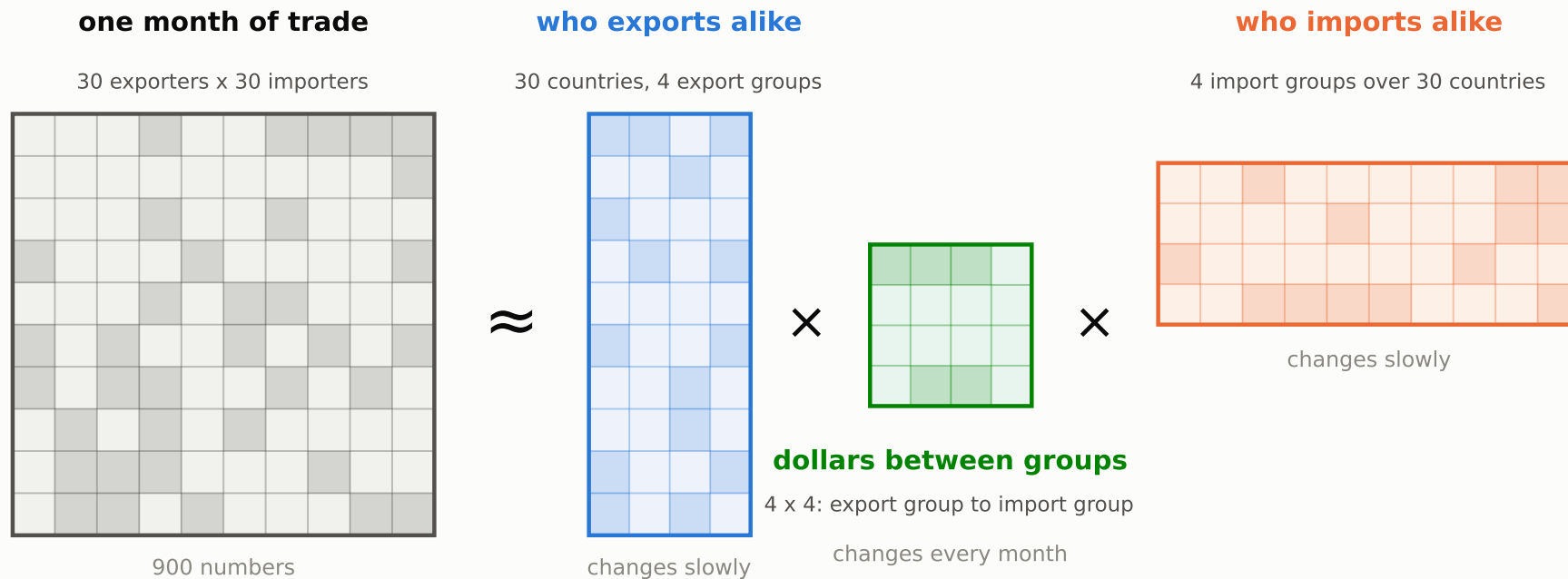
This section uses only the three blue codes (parts/GPU modules, baseboards, AI servers), monthly from 2017. Unless the page says otherwise, China and Hong Kong are counted as one. Shaded bands mark the periods between structural breaks.

How the model works

3. The factor model over time -- AI-compute codes only

The one idea behind every page that follows.

How the factor model summarizes the trade data

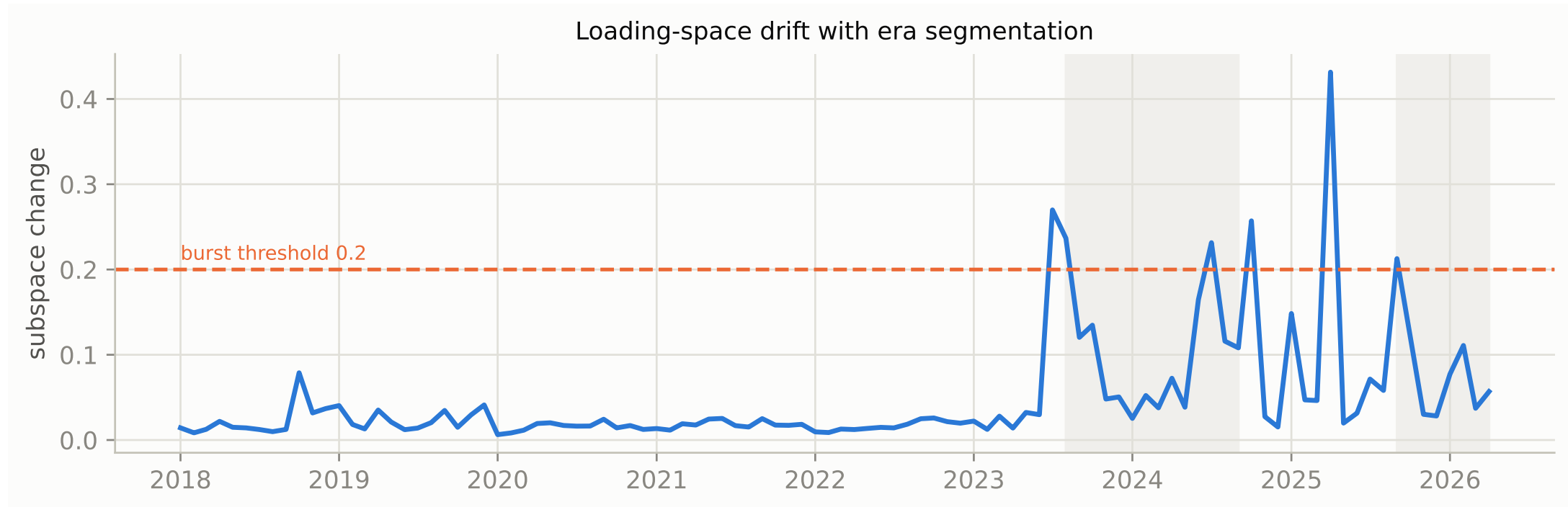


Every month, 900 country-to-country flows are compressed into: groups of exporters that move together, groups of importers, and a small table of dollars between the groups. The groups are the slow structure of the network -- when they shift, the model marks a break. The small table is the fast part: its cells are the trade channels, and three of them carry the AI boom.

When the network changed shape

3. The factor model over time -- AI-compute codes only

The line measures how much the export groupings moved from one month to the next. Flat for six years, then: Aug 2023 (AI demand), Oct 2024 (the quarter memory export controls were signaled), Sep 2025.



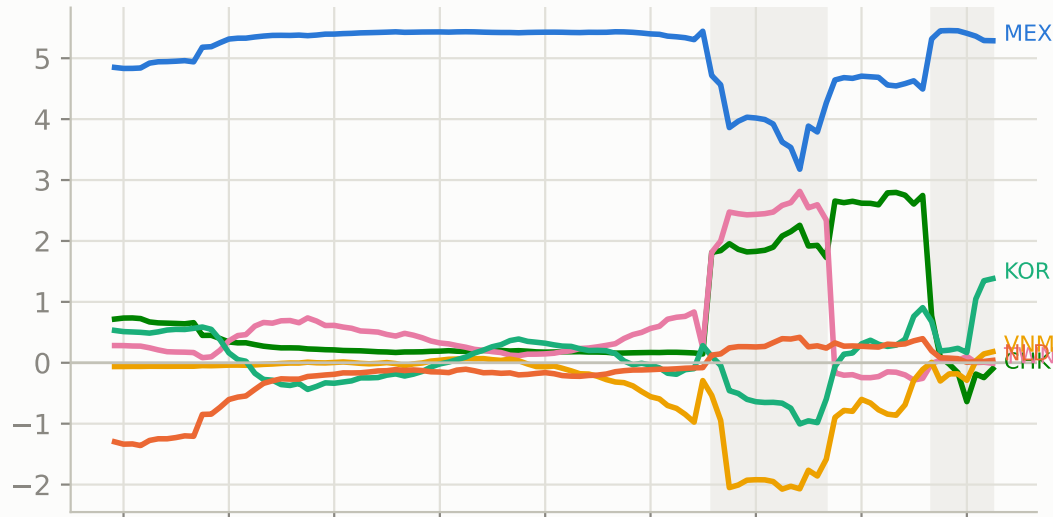
Who is in each export group

3. The factor model over time -- AI-compute codes only

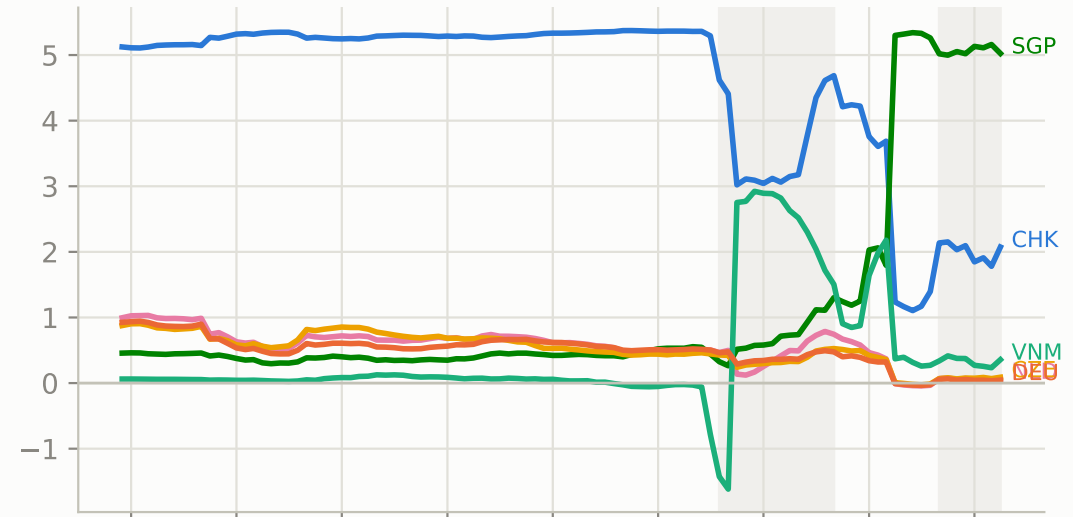
At Oct 2024 the China group loses its separate identity and a Singapore-led group appears.

AI compute, China+HKG bloc (TWN separate) — export loadings, era-anchored (12m window; shaded bands = eras)

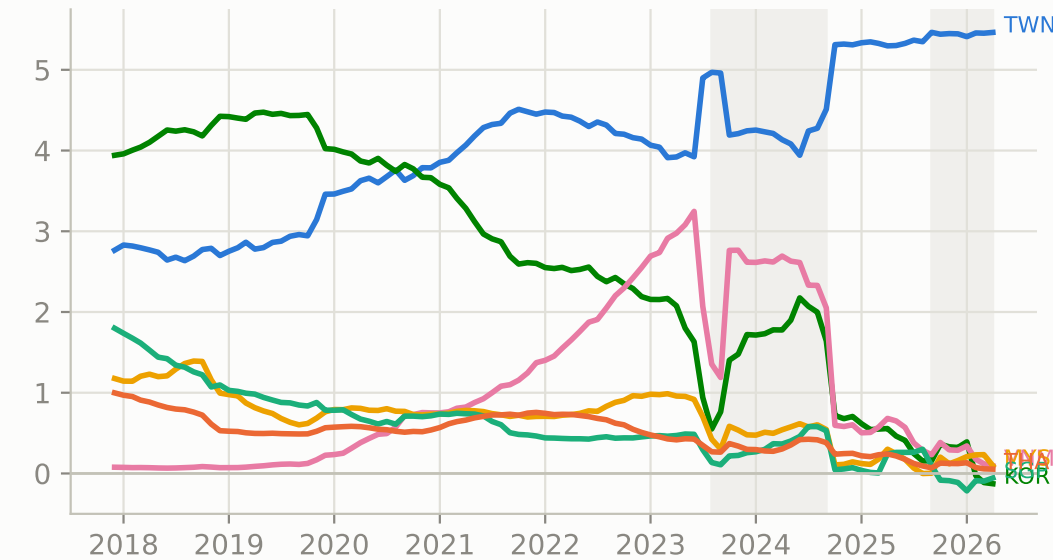
export hub 1



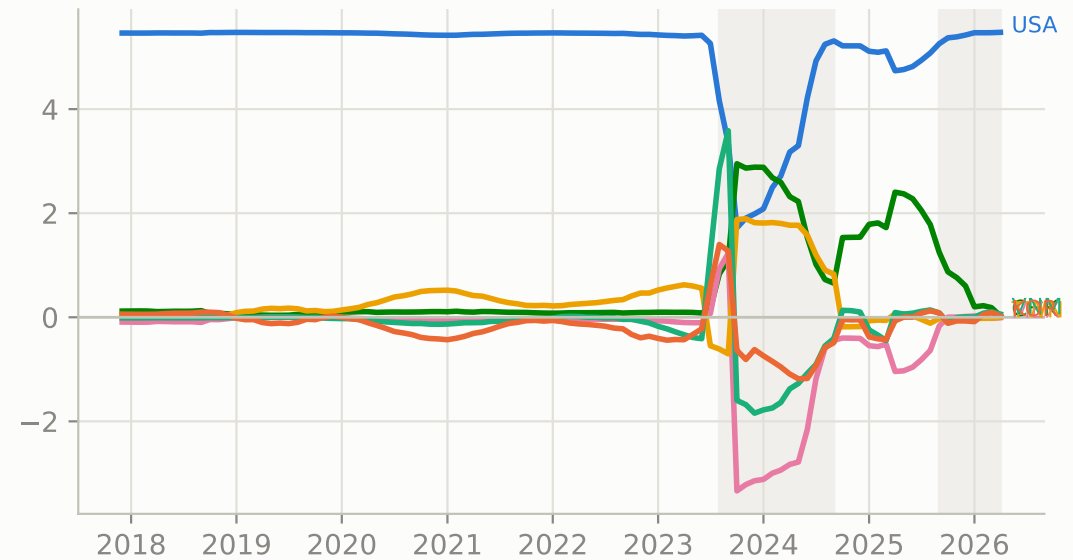
export hub 2



export hub 3



export hub 4



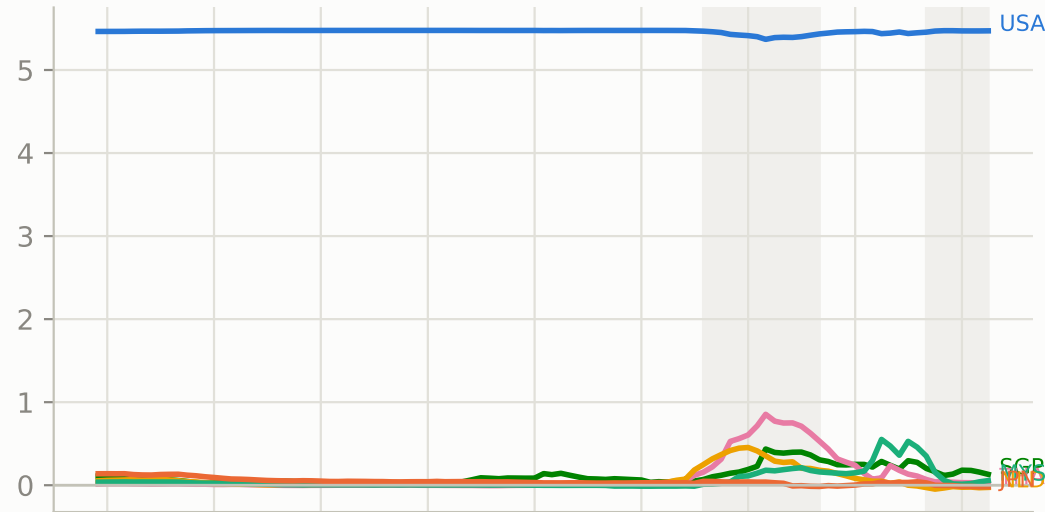
Who is in each import group

3. The factor model over time -- AI-compute codes only

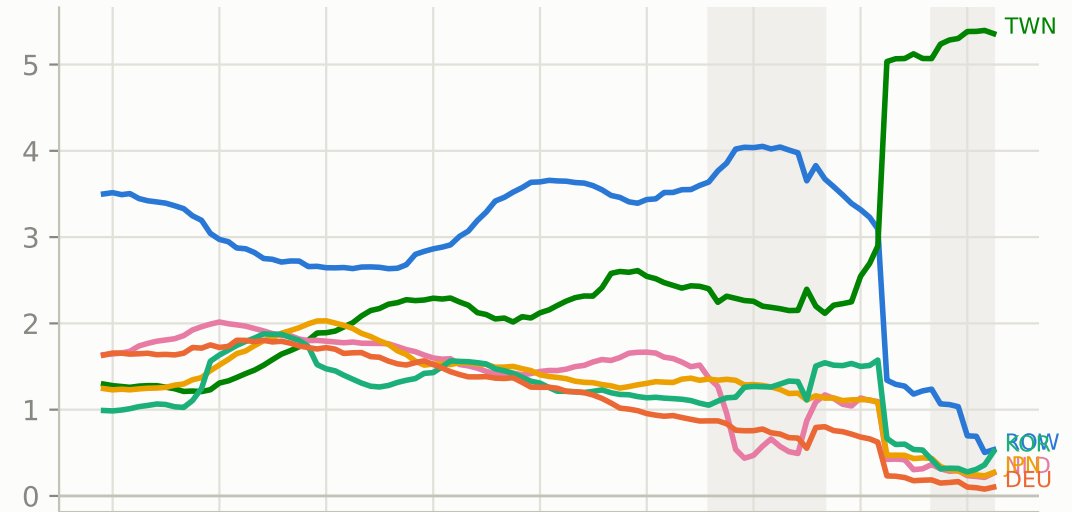
The import side is steady: a US group, a Mexico group, a Taiwan group, and the rest of the world.

AI compute, China+HKG bloc (TWN separate) — import loadings, era-anchored (12m window; shaded bands = eras)

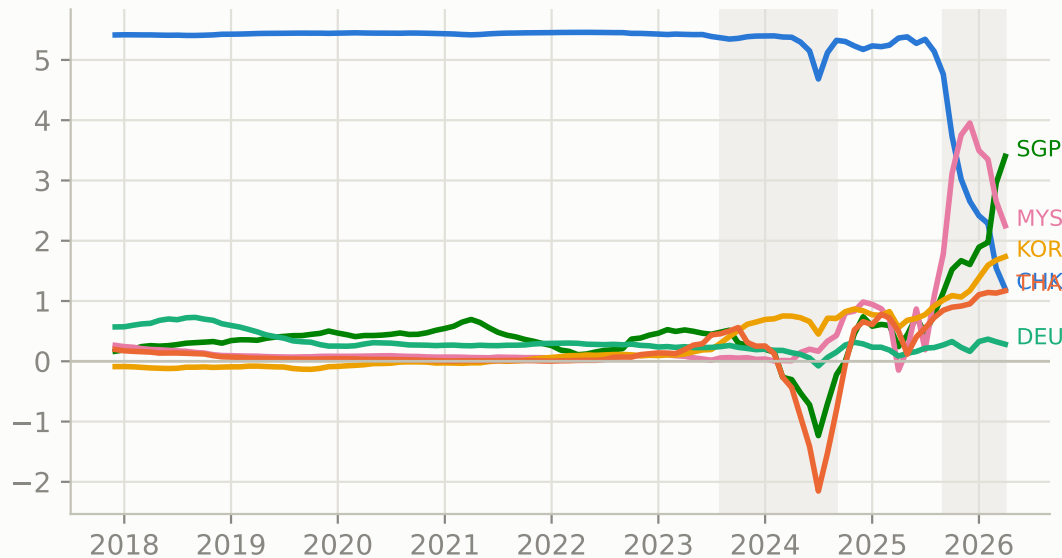
import hub 1



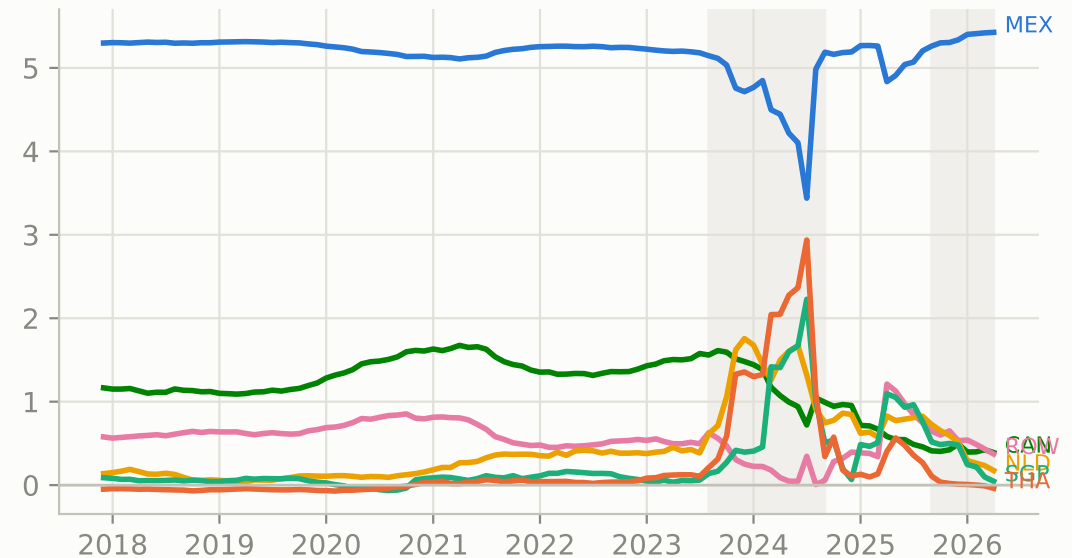
import hub 2



import hub 3



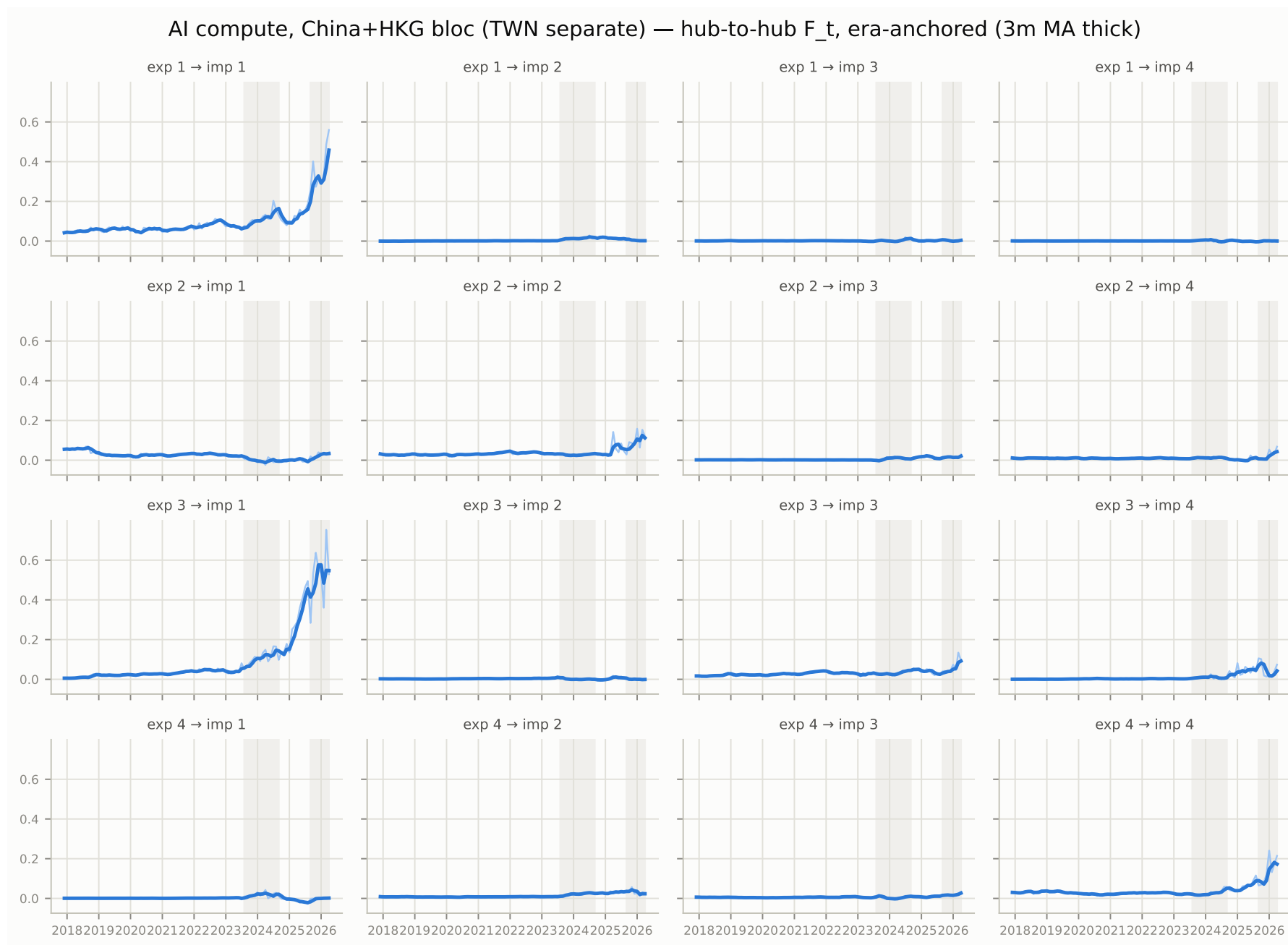
import hub 4



Dollars between groups, month by month

3. The factor model over time -- AI-compute codes only

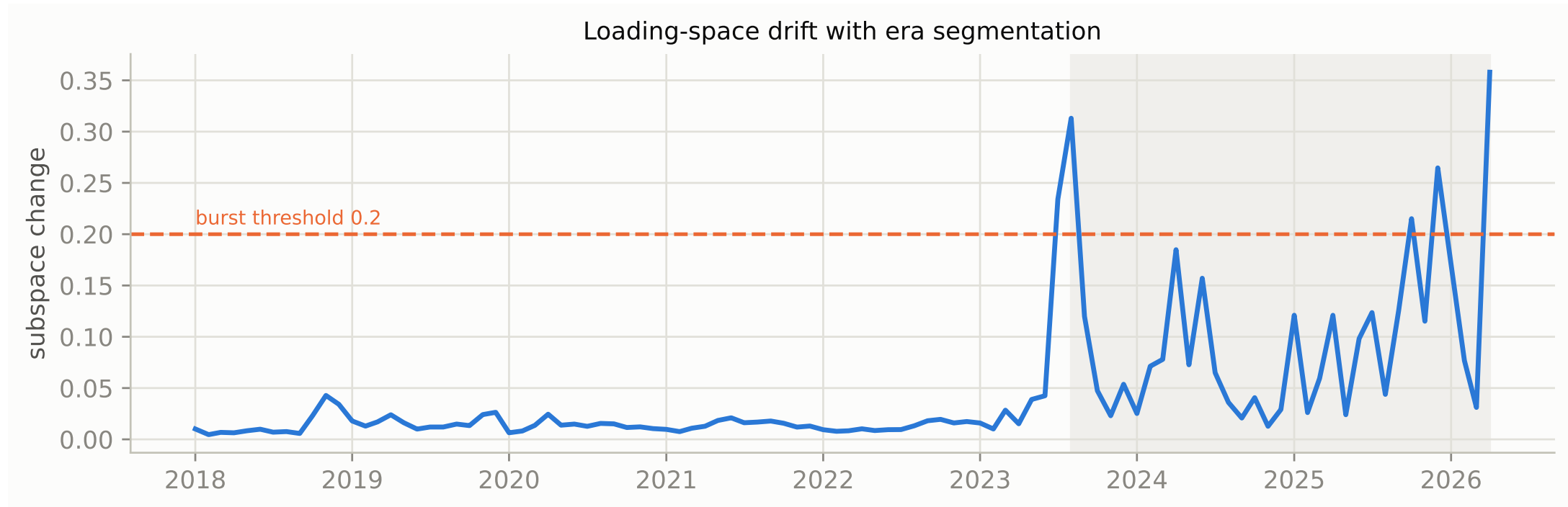
Three of the sixteen channels carry the AI boom: Taiwan to the US, and both legs of the US-Mexico assembly loop. The rest are flat.



Same, with every country separate

3. The factor model over time -- AI-compute codes only

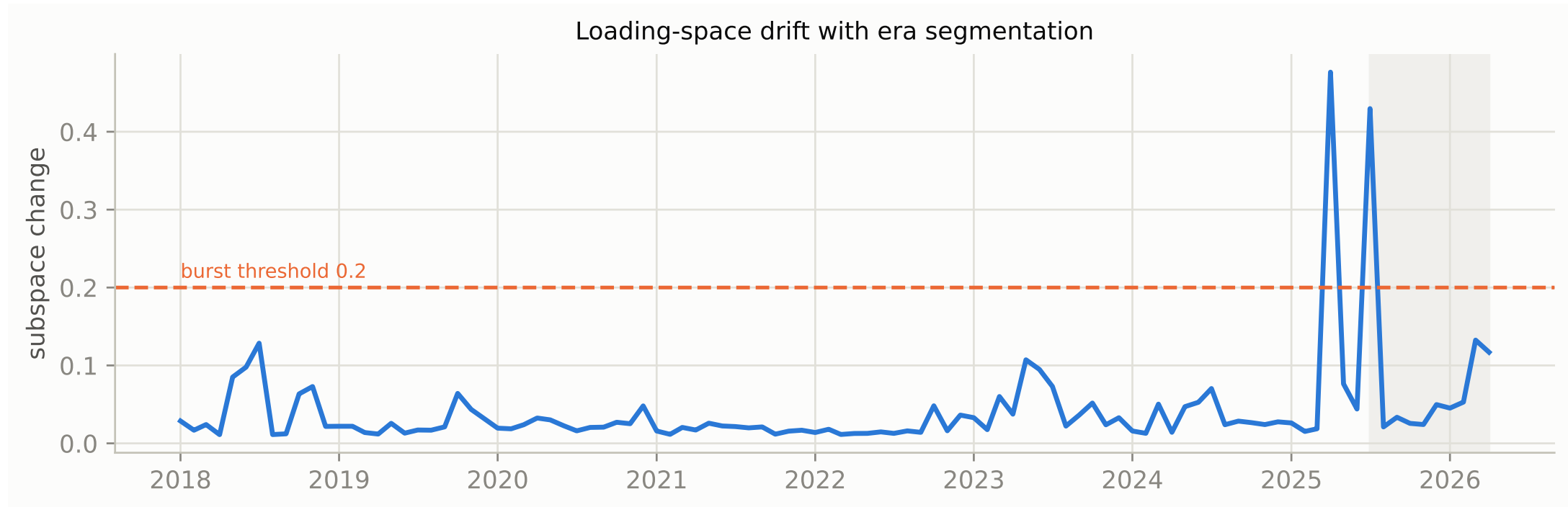
Without merging China and Hong Kong there is one break in nine years: Aug 2023. Demand changed the country-level structure; policy did not.



Parts only (847330)

3. The factor model over time -- AI-compute codes only

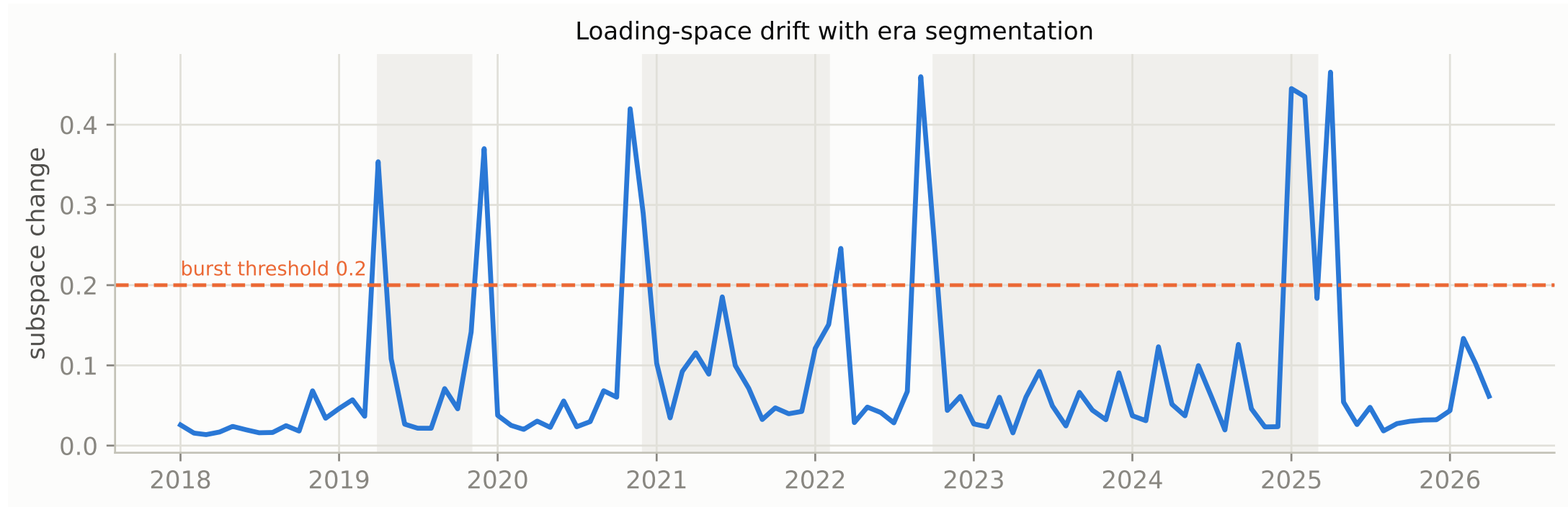
91 months without a break -- the stable base layer of the network.



With both blocs merged

3. The factor model over time -- AI-compute codes only

Merging China+Hong Kong and US+Mexico hides churn inside each bloc and isolates what happens between them. The breaks line up with policy: the 2019 tariff war, the Oct 2022 export controls, the Apr 2025 tariffs.



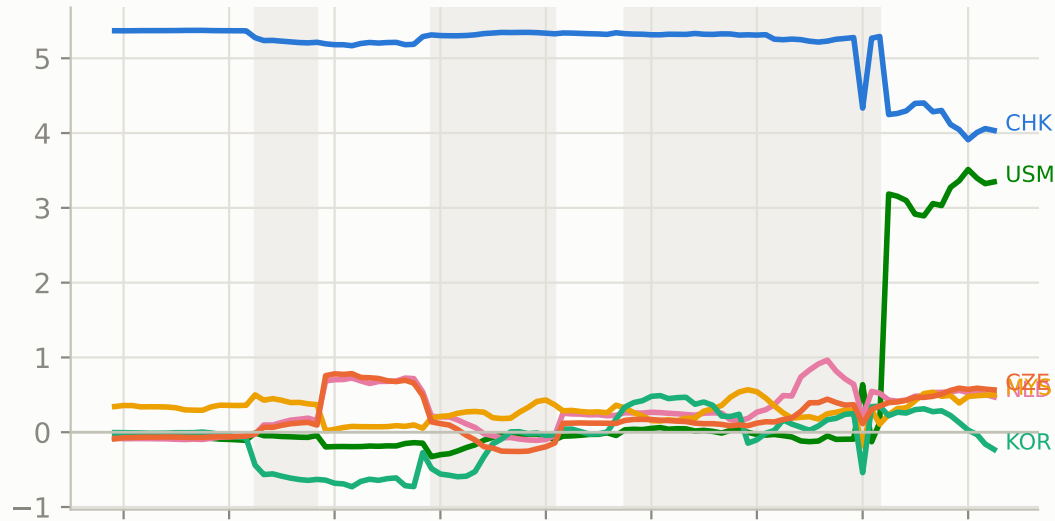
The tariff signature

3. The factor model over time -- AI-compute codes only

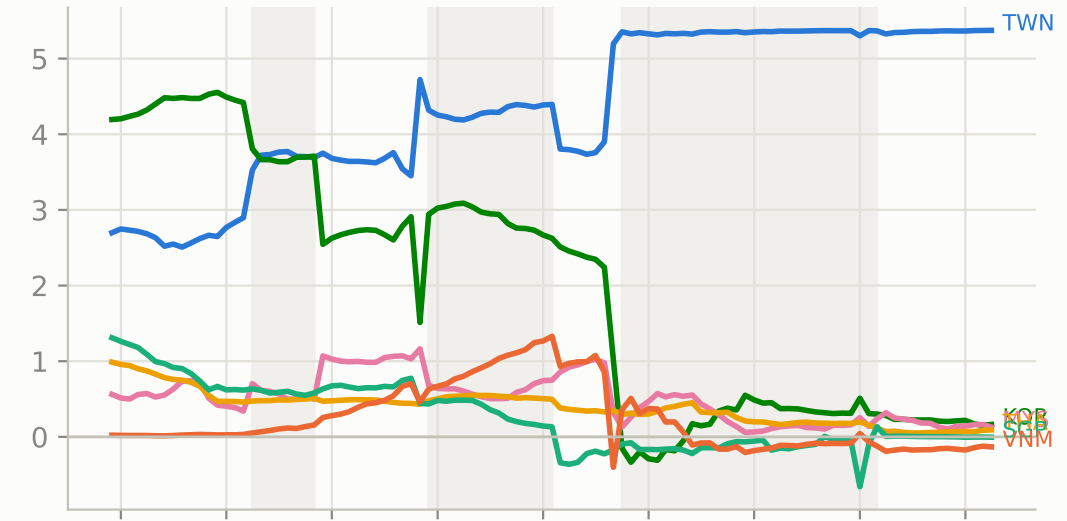
After Apr 2025 the two blocs move together on a single factor -- seen at no other break.

AI compute, CHN+HKG and USA+MEX blocs — export loadings, era-anchored (12m window; shaded bands = eras)

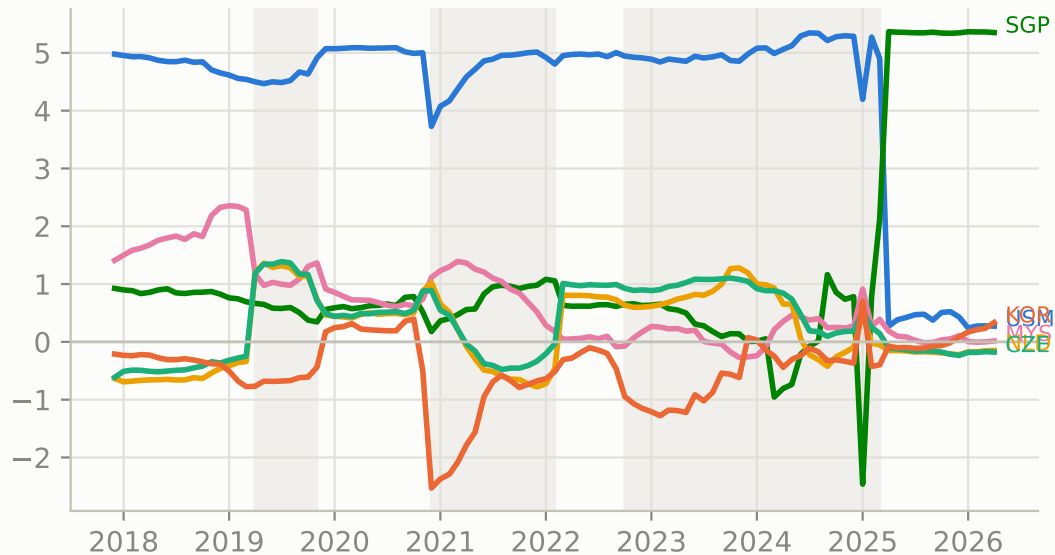
export hub 1



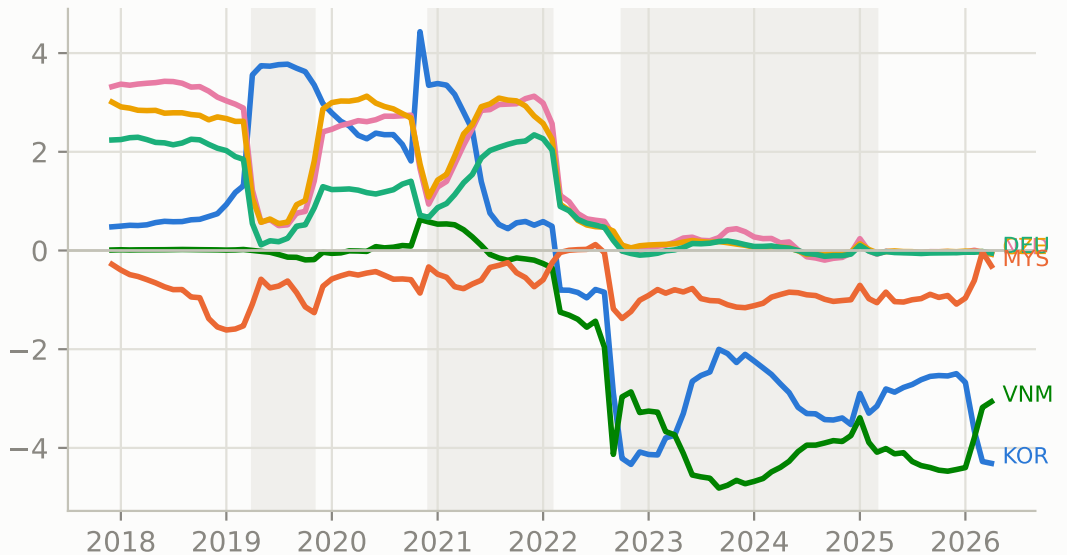
export hub 2



export hub 3



export hub 4



4. Concentration & fragmentation -- AI-compute codes only

Monthly measures built on top of the model, for the three blue codes only. Dotted lines mark Jul 2023 and Apr 2025. Interpretation: [results/network_stats/findings.md](#).

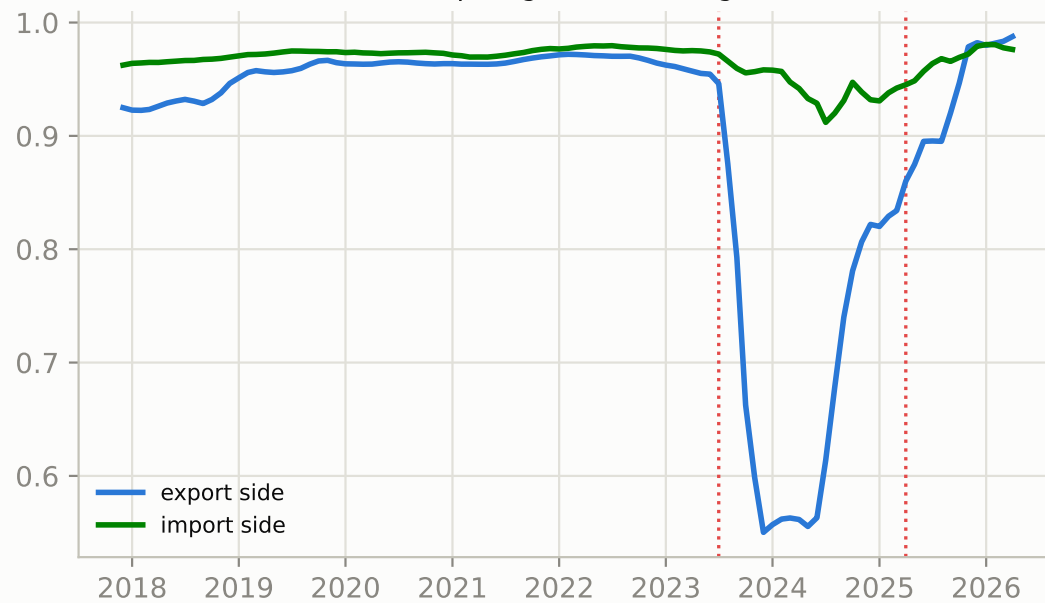
Two meanings of fragmentation

4. Concentration & fragmentation -- AI-compute codes only

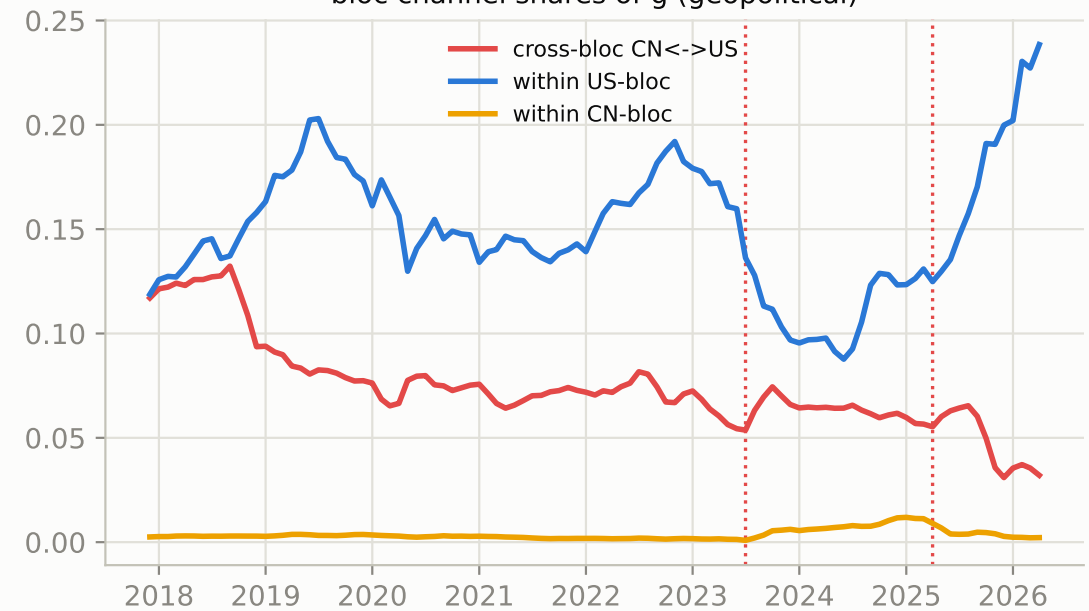
Left: do the model's groups overlap (a technical check, little economic signal). Right: the share of trade crossing between the China and US blocs -- down about 70% since 2017, in two steps (2019, then 2023 on), with no recovery between.

AI compute (3 codes) — two notions of fragmentation (3m MA)

hub-overlap fragmentation (agnostic)



bloc channel shares of g (geopolitical)



Concentration, taken apart

4. Concentration & fragmentation -- AI-compute codes only

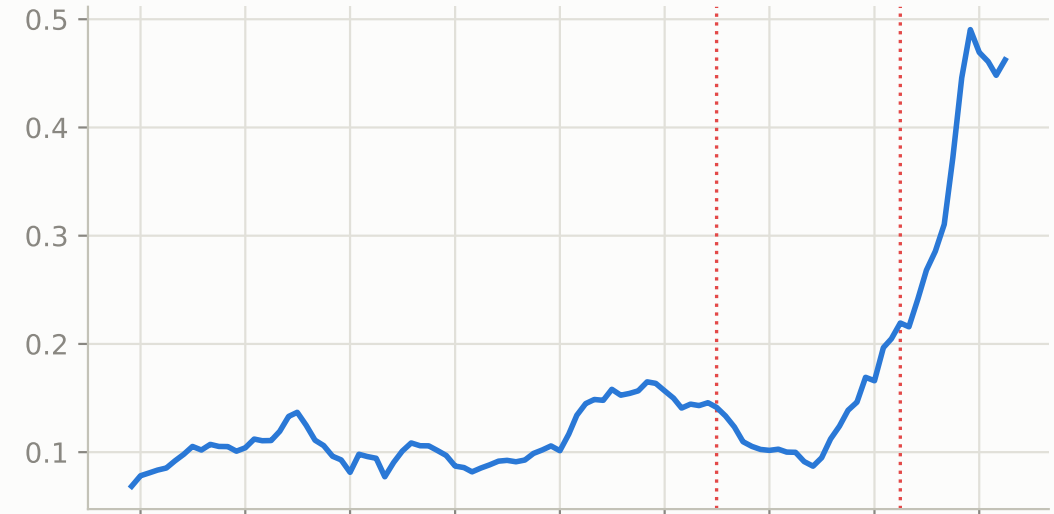
Overall concentration splits exactly into three parts. The interesting one asks: do concentrated sellers ship to concentrated buyers? Mildly yes, peaking around 2019, fading through the AI era.

AI compute (3 codes) — joint concentration decomposition (3m MA; dotted = 2023-07, 2025-04)

linkage concentration HHI_F



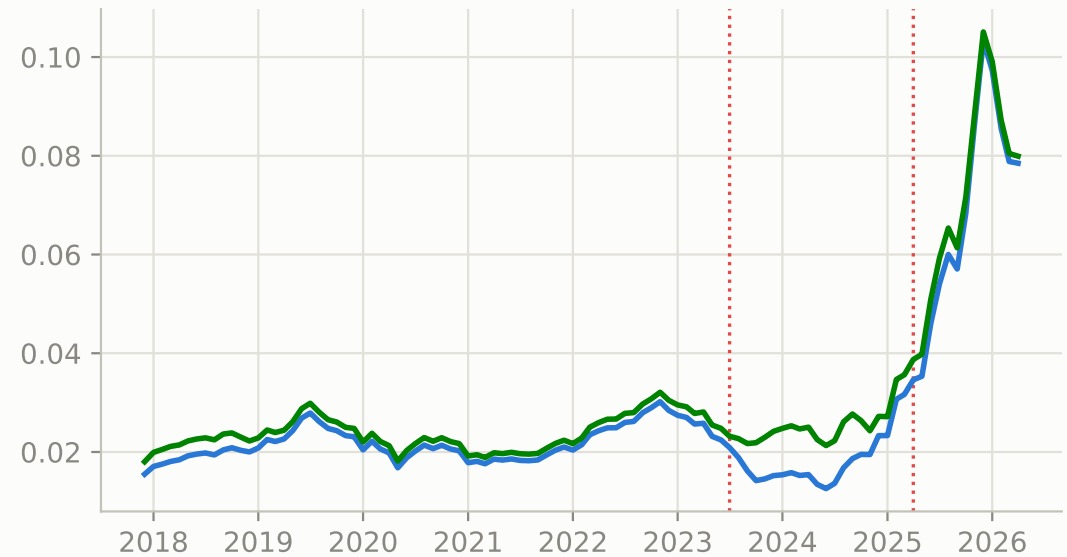
linkage-weighted means $\mu_a * \mu_b$



alignment covariance $\text{Cov}(a,b)$



joint: HHI_flat (blue) vs bilateral (green)

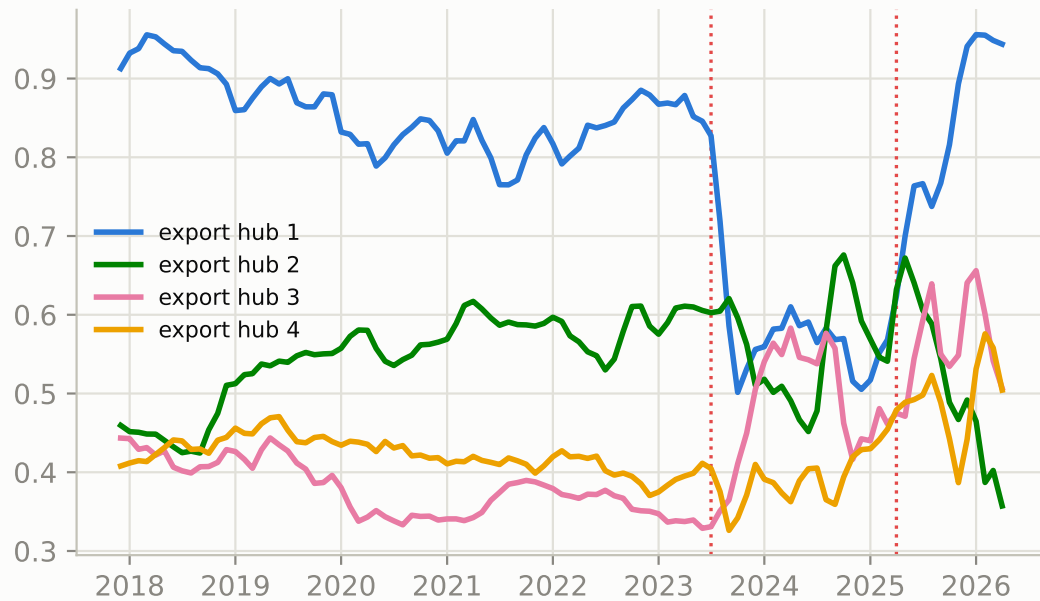


One buyer or many?

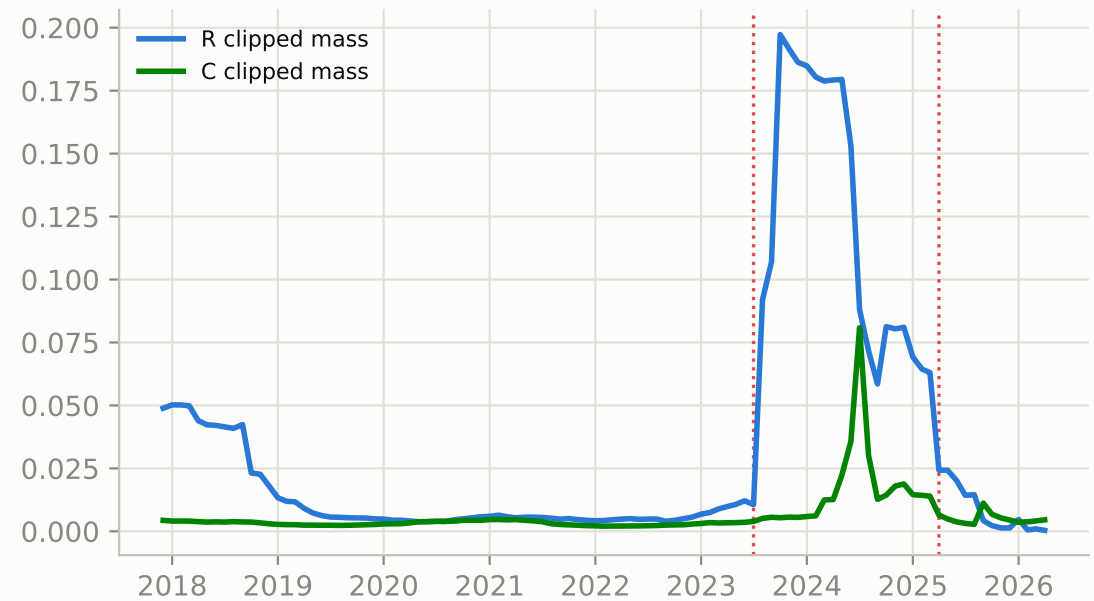
Left: does each export group sell into one import group or several. Right: a data-quality check, elevated during the 2023-24 transition.

AI compute (3 codes) — linkage structure and basis diagnostics

per-hub linkage HHI (1 = sells to a single import hub)



negative mass clipped (basis-quality diagnostic)

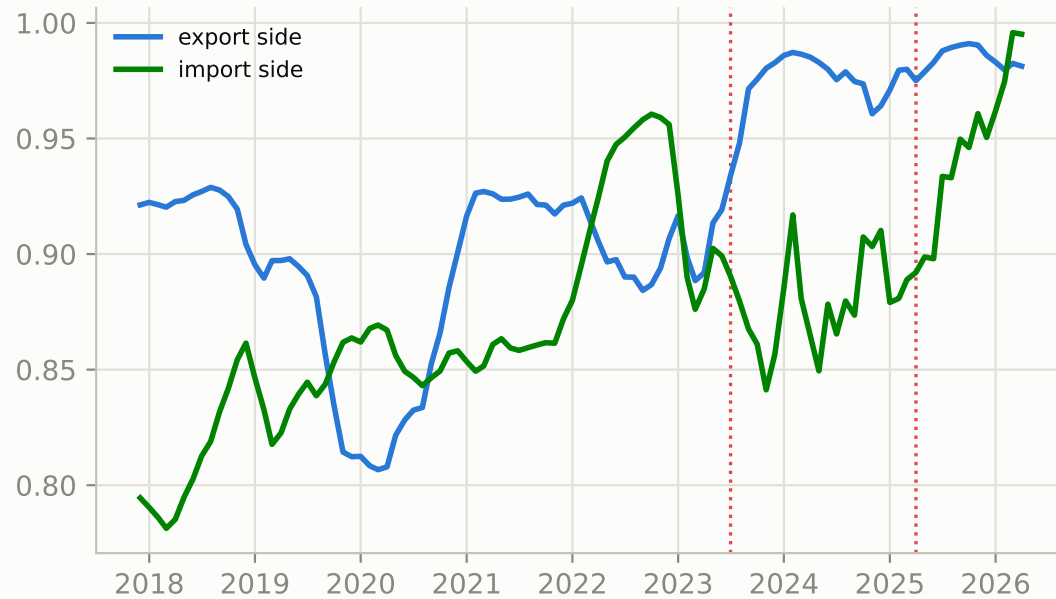


Baseboards: the sharpest split

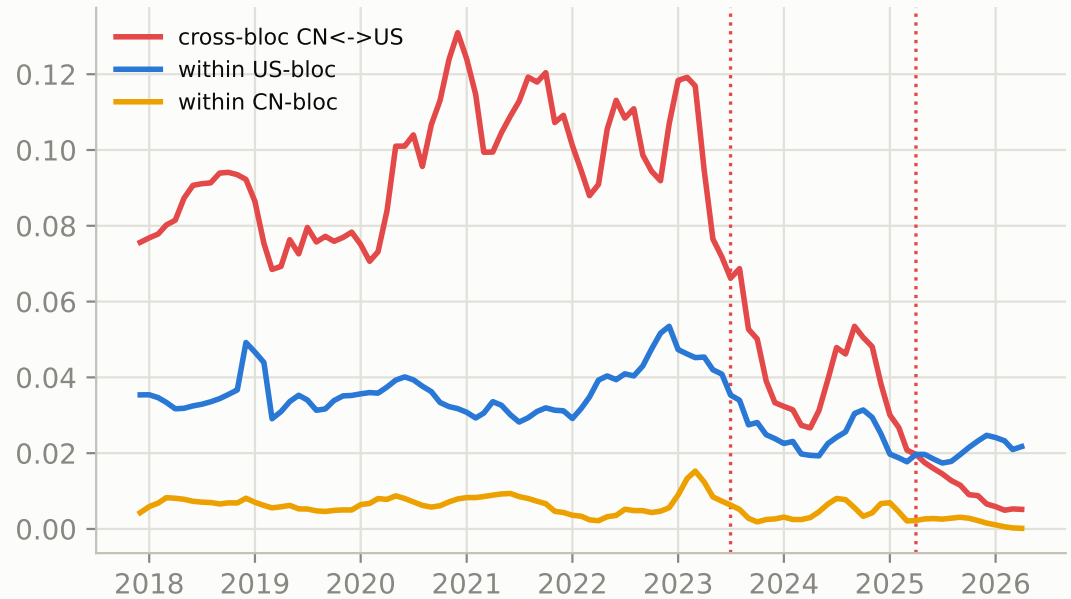
The cross-bloc share collapses about 20x from its 2021 peak -- the code most exposed to export controls.

HS 847180 (baseboards) — two notions of fragmentation (3m MA)

hub-overlap fragmentation (agnostic)



bloc channel shares of g (geopolitical)

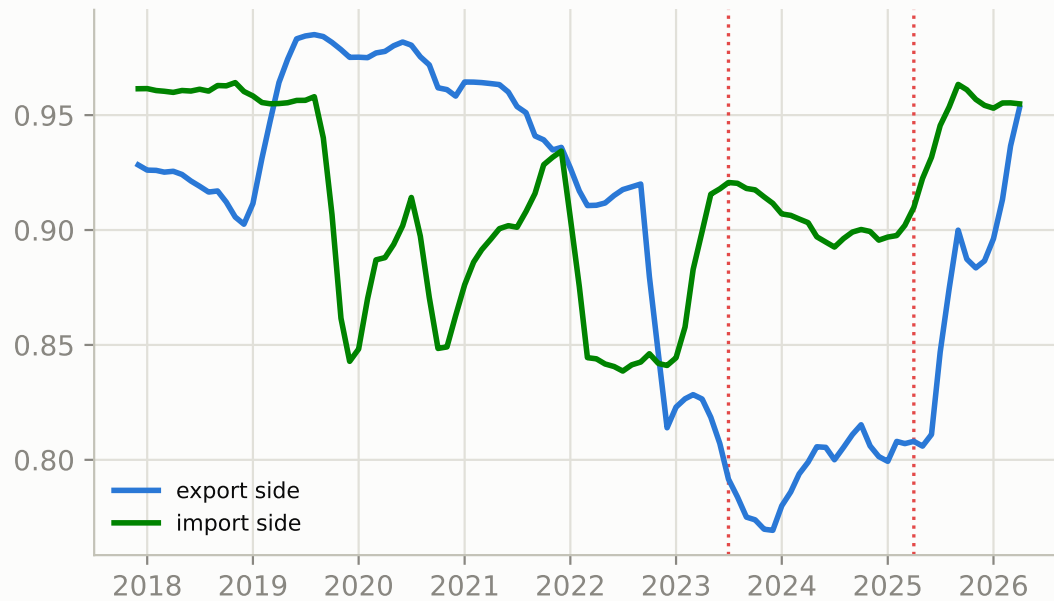


Parts: still shared

The cross-bloc share halves but stays the highest of the three codes: the parts trade remains the layer both blocs still share.

HS 847330 (parts/GPU cards) — two notions of fragmentation (3m MA)

hub-overlap fragmentation (agnostic)



bloc channel shares of g (geopolitical)

